Coupling biofilm growth into an existing FEM element

Master's thesis colloquium

Keisuke Nishioka

Institute of Continuum Mechanics (IKM), Leibniz Universität Hannover

Examiners: Prof. Dr.-Ing. Philipp Junker, Dr.-Ing. Matthias Wangenheim

Supervisors: Dr.-Ing. Meisam Soleimani, Dr.-Ing. Hendrik Geisler

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Motivation and objective

- Peri-implantitis affects about one in five implant patients (prevalence 22 % and 19.5 % in two meta-analyses)
- It is driven by multi-species biofilms; a shift towards pathobionts such as *P. gingivalis* goes with disease
- A biofilm is also a mechanical body: where it grows unevenly or against a stiff substrate, it carries stresses

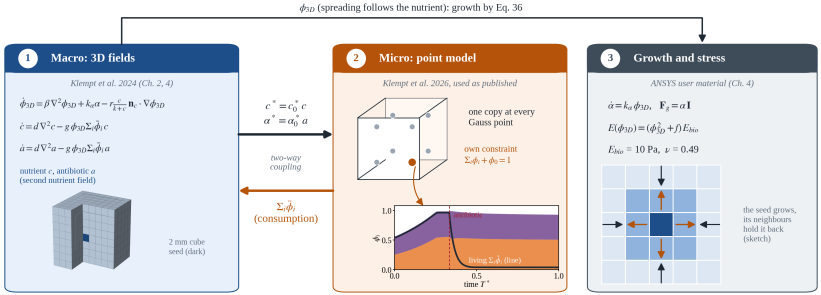
Objective: to compute the species composition of a biofilm, its growth and the resulting stresses consistently within an existing finite element framework.

Point model
species composition at a material point
no spatial resolution, no stress

Growth field
spatial growth and growth-induced stress
single biofilm phase, no species

This work
both, evaluated at every Gauss point of an existing ANSYS element

Approach



Output: ϕ_i, ψ_i at each Gauss point. $\bar{\phi}_i = \phi_i \psi_i$: only living biomass consumes. Assumed, not from a paper: c^*, a^* linear in time, $\Sigma_i \bar{\phi}_i$ one step late, new points take the neighbours' living composition, $\phi_{min} = 0.001$ (converged). x from the data window.

1. Calibration
which of the 15 interactions the data determine (TMCMC, 4 conditions, 3 seeds)

2. Growth law in ANSYS
$\mathbf{F}_g = \alpha \mathbf{I}$ as a user material, verified against an exact solution

3. Coupling
composition reaches the stress only through nutrient and front speed; grid error known

$\rightarrow$ The point model runs unchanged at every Gauss point; nutrient and antibiotic come from the fields, and only living biomass consumes them.

Both models from the extended Hamilton principle

Growth field (Klempt et al. 2024)

$$\dot{\phi} = \beta \nabla^2 \phi + k_\alpha \alpha - r \tfrac{c}{k+c} \mathbf{n}_c \cdot \nabla \phi$$
$$\dot{c} = d \nabla^2 c - g\, \phi \sum_i \bar{\phi}_i\, c, \qquad \dot{\alpha} = k_\alpha \phi$$
$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g, \quad \mathbf{F}_g = \alpha \mathbf{I}, \quad \alpha(0) = 1$$

Stress only from $\mathbf{F}_e$; growth is $\alpha - 1$.

$\phi \in [0,1]$: amount of biofilm; c: nutrient concentration; α: growth variable, growth $= \alpha - 1$; β: diffusion coefficient of ϕ [mm$^2/T^*$]; k_α: growth rate [$1/T^*$]; d, g: nutrient diffusivity [mm$^2/T^*$], consumption rate [$1/T^*$]; T^*: normalised time of Klempt et al. 2024

Point model (Klempt et al. 2026)

$$\Psi + \Delta + C \;\rightarrow\; \underset{\dot{\boldsymbol{\phi}}, \dot{\boldsymbol{\psi}}, \gamma}{\text{stat}}$$

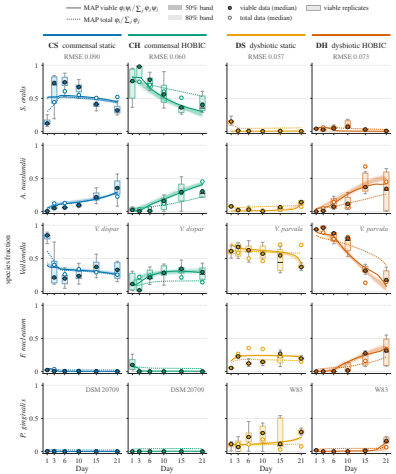
$\Psi = -\frac{1}{2} c^* \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i \psi_i$, $C = \gamma(\sum_l \phi_l - 1)$

local form: principle of the minimum of the dissipation potential (Junker et al. 2014, Hackl 1997)

$\Rightarrow$ $\mathbf{A}$ symmetric, $2n+2$ equations per point

ϕ_i, ψ_i: volume fraction and viability of species i; $\bar{\phi}_i$: living fraction; Ψ: free energy, Δ: dissipation function, C: constraint function; c^*: nutrient level; $\mathbf{A}$: interaction matrix; γ: Lagrange multiplier of $\sum_{l=0}^{n} \phi_l = 1$ (ϕ_0: void)

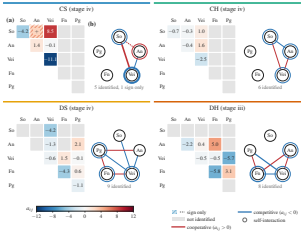
Part 1: Bayesian calibration of the point model

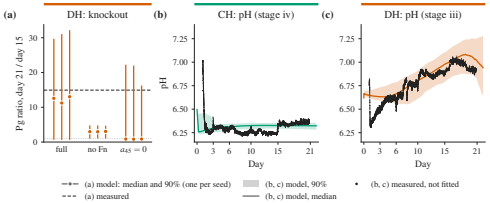


- Heine et al. (2025): five species, four conditions (commensal / dysbiotic, static / flow), days 1–21
- 15 coefficients a_{ij} per condition, effective and undirected
- TMCMC in four stages, three seeds each, with convergence checks
- JAX on the GPU: about 150 times faster than the CPU

Fit: RMSE 0.03 to 0.18; in CS composition and live fraction cannot both be matched

Part 1: what the data determine

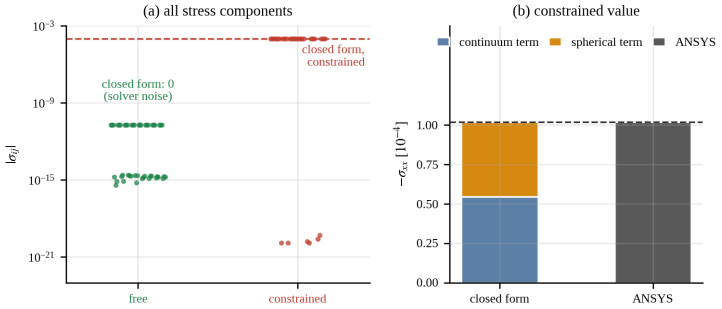




- Identified entries: 4, 6, 8, 9 of 15 (CS, CH, DS, DH)
- Commensal: only the early colonisers; *So–Vei* positive in both
- pH not used in the inference: a check, followed in DH

→ **Testable hypothesis:** without *F. nucleatum*, the late growth of *P. gingivalis* under dysbiotic flow disappears in the model.

Part 2: implementation of the growth law in ANSYS



Material

ANSYS user material, $\mathbf{F}_g = \alpha\mathbf{I}$, neo-Hookean

Stiffness (Klempt et al. 2024)

$E(\phi) = (\phi^2 + f)E$, $E = 10$ Pa, $\nu = 0.49$; $f = 10^{-3}$ (this work)

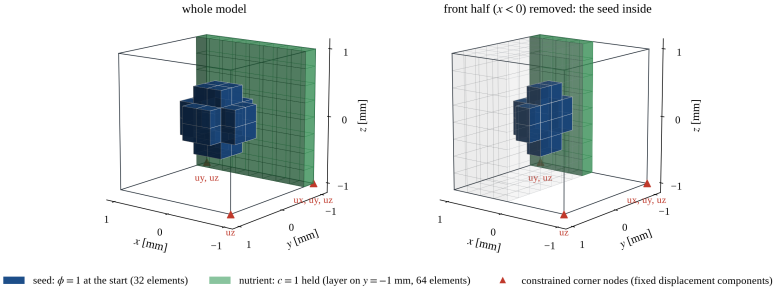
Check (one element)

free: zero stress; constrained: -1.0193×10^{-4} MPa, as predicted

$\rightarrow$ The implementation reproduces the closed-form solutions for free (stress-free) and constrained growth.

Part 2: coupling to the IKM element

ANSYS model: cube of 2 mm, $8 \times 8 \times 8 = 512$ hexahedral elements of 0.25 mm



Model

2 mm cube, 8^3 SOLID185 (B-bar), seed of 32 elements; also 16^3, 24^3

Unchanged

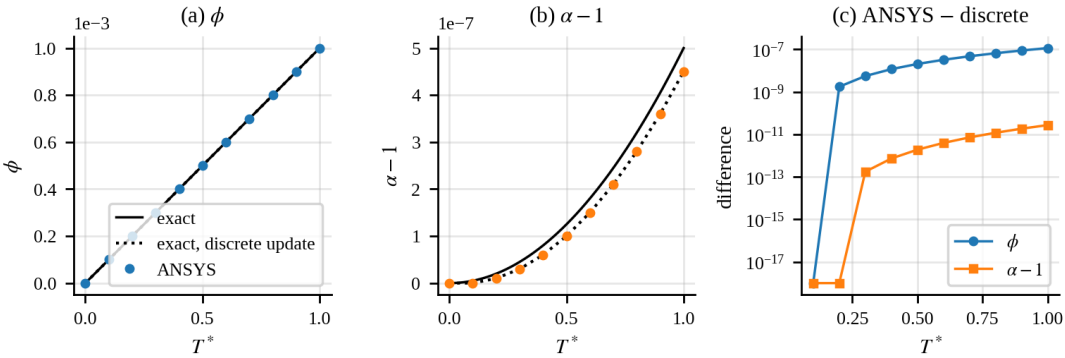
the element's field ϕ (NEM); $\beta = 0.02$ mm$^2/T^*$, converted (assumption)

Added in this work

$\dot{\alpha} = k_\alpha \phi$ at every Gauss point (Eq. 36) $\rightarrow \mathbf{F}_g \rightarrow$ stress

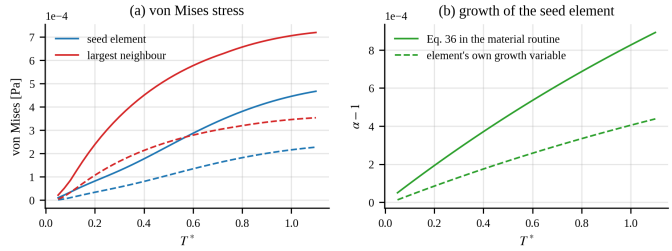
$\rightarrow$ The field formulation of the element is retained; only the growth law at the Gauss points is added.

Part 2: verification against an exact solution



$\rightarrow$ Where ϕ has no gradient, $\phi = \sinh(k_\alpha t)$ and $\alpha - 1 = \cosh(k_\alpha t) - 1$: the coupled ANSYS solution agrees with the exact solution of the time integration to 3×10^{-11} in α.

Part 2: growth-induced stresses with the published stiffness



Left: von Mises stress of the seeded element and its largest neighbour; right: $\alpha - 1$; solid: Eq. 36 at the Gauss points, dashed: the element's own growth variable.

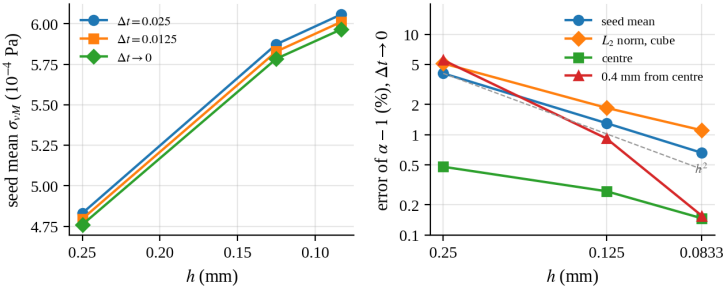
At $T^* = 1.1$ on 16^3 elements:

mean stress, seed	-2.5×10^{-4} Pa
mean stress, first layer	$+6.0 \times 10^{-5}$ Pa
mean stress, second layer	$+1.3 \times 10^{-6}$ Pa
von Mises, seed mean	5.9×10^{-4} Pa
$\alpha - 1$ in the seed	8.9×10^{-4}

Stresses scale with $E = 10$ Pa.

$\rightarrow$ The seed is under compression and the surrounding material under tension, consistent with the ring of tension reported by Klempt et al. 2024.

Part 2: convergence and stability



Time step Δt [T^*]

error about 1 %

Element size h [mm]

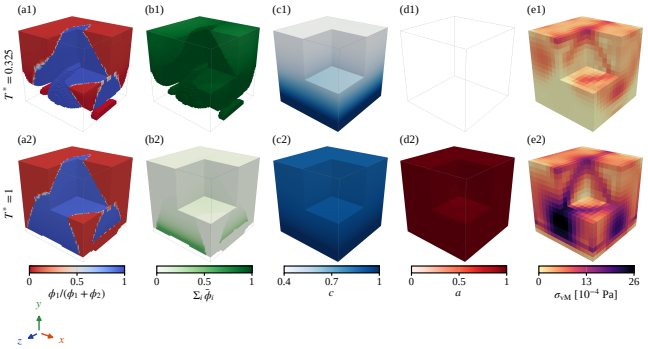
error of $\alpha - 1$ measured against a reference: 0.7 % on 24^3

Stability

limit of the element derived and confirmed; all runs below it (Appendix D)

$\rightarrow$ The errors are measured, not estimated, and every run lies within the derived stability limit.

Part 3: two species with nutrient and antibiotic

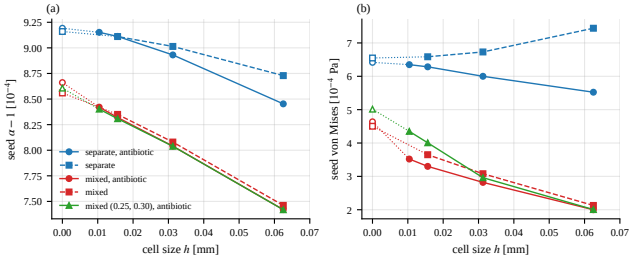


Separate start, $T^* = 0.325$ (top) and 1 (bottom): share of A, $\sum_i \bar{\phi}_i$, c, a, von Mises stress; 64^3, one octant removed

- Synthetic data (Fritsch et al. 2025, case I), all constants from the papers; antibiotic from $T^* = 1/3$, $\alpha_0^* = 100$
- Living fraction of the seed: separate $0.96 \to 0.15$, mixed $0.96 \to 0.04$
- Growth changes by less than 2 % and the converged seed stress by about 2 %: the antibiotic lowers the living fraction, not ϕ_{3D}
- New points take the living composition of their neighbours (assumption)

$\to$ The composition reaches growth and stress only through the nutrient and the front speed $rc/(k+c)$.

Part 3: the front term sets the accuracy of the stress

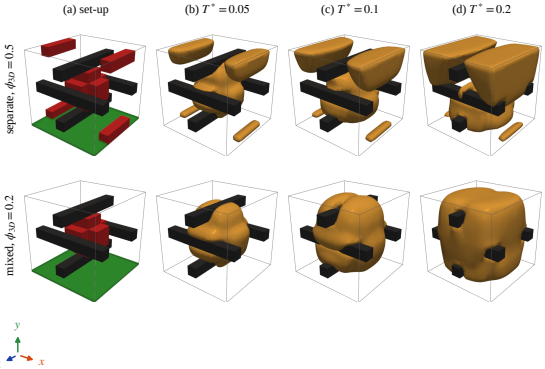


Seed means at $T^* = 1$ against the cell size, all examples, outer step 0.0025; open symbols: extrapolation (three finest grids, Celik et al. 2008; 192^3 with antibiotic)

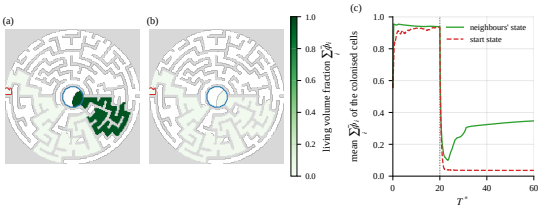
- Front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$: cell Péclet number ≈ 31 on 16^3; a constant added diffusion gives a quarter of the converged stress, a second-order upwind scheme is needed
- With it, up to 192^3: growth GCI below 4.5 %, separate stress 1.4 %; mixed stress 19 to 41 % with minmod, 4.6 % with the superbee limiter, which keeps the edge sharper (Backup 17)
- The IKM element with upwind gradients: 5.8×10^{-4} Pa (GCI 16.5 %), inside the band

→ The stress needs a second-order treatment of the front term; with it the grid error is known.

Part 3: rigid bars and a maze



Four rigid bars after Klempt et al. 2024, Fig. 9; surface ϕ_{3D} at $T^* = 0.05$ to 0.2



Maze of Klempt et al. 2024, Fig. 11, antibiotic from $T^* = 20$: new points take the neighbours' state (a) or the start state (b); living fraction over time (c)

- Bars restrain the swelling: seed von Mises about ten times higher (32^3, separate start)
- Maze: the rule for new points decides recovery after the antibiotic, 0.35 against 0.04 (assumption)

→ Geometry changes the stress; the start state of new points decides whether the biofilm recovers.

Limitations

- **Verification, not validation**: no measured stress of a growing oral biofilm was available
- **Mechanics one way**: the stress does not act back on growth (as in Klempt et al. 2024), and the composition does not act on the stiffness
- **Front term** $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$: its speed depends on the nutrient only, so dead biofilm keeps spreading after the antibiotic; the stiffness counts dead biomass
- **Assumptions**: start values of newly reached points, $\phi_{\min}$, the conversion of β; each reported with its sensitivity
- **Calibrated parameters in the FE model**: one earlier one-way run (CH) only; Part 3 uses synthetic data
- **Symmetric A**: no directed effects; only part of $\mathbf{A}$ is determined; different strains in the commensal and dysbiotic data

$\rightarrow$ Each limitation is quantified in the thesis where possible; the results are verified, not yet validated.

Conclusions and outlook

Contributions

1 Calibration: five species, four conditions; which interactions the data determine

2 Growth law: Klempt et al. 2024 in the IKM element, verified to 3×10^{-11}

3 Composition: the point model at every Gauss point, coupled through nutrient and antibiotic; it reaches the stress only through the nutrient and the front speed

Outlook (continuation at Keio University)

- Five calibrated species in the finite element model
- Stress into growth (Eq. 30 of Klempt et al. 2024), composition into stiffness
- A second-order front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$ in the element; explicit against implicit field update
- Posterior uncertainty carried to the stress

References

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- F. Klempt, H. Geisler, M. Soleimani, P. Junker (2026), Arch. Appl. Mech. 96, 164: multi-species point model
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- T. Rudolf et al. (2025), Finite Elem. Anal. Des. 249, 104353: Neighbored Element Method
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Appendix A: notation

Growth field and mechanics (Klempt et al. 2024)

ϕ, ϕ_{3D}	amount of biofilm (0 = void, 1 = dense)
c	nutrient concentration
α	$\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$
$\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$	elastic / growth split
k_α	growth rate ($10^{-3}\, T^{*-1}$)
β	diffusion of ϕ (0.02 mm$^2/T^*$, converted)
d, g	nutrient diffusivity, consumption
r, k	front growth parameter, half-velocity constant
E, ν	10 Pa, 0.49 (Klempt 2024, Table 2)
T^*	normalised time

Point model (Klempt et al. 2026)

ϕ_i, ϕ_0	volume fraction of species i, void
ψ_i, $\bar{\phi}_i$	viability, living fraction $\phi_i\psi_i$
A_{ij}	symmetric interaction matrix
c^*, α^*	nutrient, antibiotic level
η_i	dissipation parameters
γ	multiplier of $\sum_l \phi_l = 1$

Coupling (this work)

$\phi_1/(\phi_1 + \phi_2)$	composition (share of species 1)
s	point model advances $s\,\Delta t$ per substep
ϕ_{cap}, ϕ_{min}	cap on the amount, start threshold
$\lambda = \beta\Delta t/h^2$	stability measure

Appendix B: growth field and mechanics

Kinematics. $\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$ with isotropic $\mathbf{F}_g = \alpha \mathbf{I}$. Stress comes only from $\mathbf{F}_e$. A body that can grow freely stays stress free; stress appears where growth is constrained.

Material. $E(\phi) = (\phi^2 + f)\, E$, $E = 10$ Pa, $\nu = 0.49$ (Klempt 2024, Table 2; their material is incompressible), $f = 10^{-3}$; decks in mm and MPa, stresses quoted in Pa.

Field equations (Klempt et al. 2024, Eqs. 34–36, without the bars):

$$\dot{\phi} = \underbrace{\beta \nabla^2 \phi}_{\text{diffusion}} + \underbrace{k_\alpha\, \alpha}_{\text{local source}} \underbrace{- \|\nabla \phi\| \frac{r\, c}{k + c} \mathbf{n}_{\nabla\phi} \cdot \mathbf{n}_{\nabla c}}_{\text{front, inactive in ANSYS}}, \qquad \dot{c} = d \nabla^2 c - g\, \phi\, c, \qquad \dot{\alpha} = k_\alpha\, \phi$$

- ϕ and c are fields of the element, updated explicitly with the Neighbored Element Method; α is a state variable at the Gauss points
- Consumption $g\phi c$ as in the authors' implementation (printed: $g\phi$)

Appendix C: point model and one coupled substep

Point model (Klempt et al. 2026), per species i, no space:

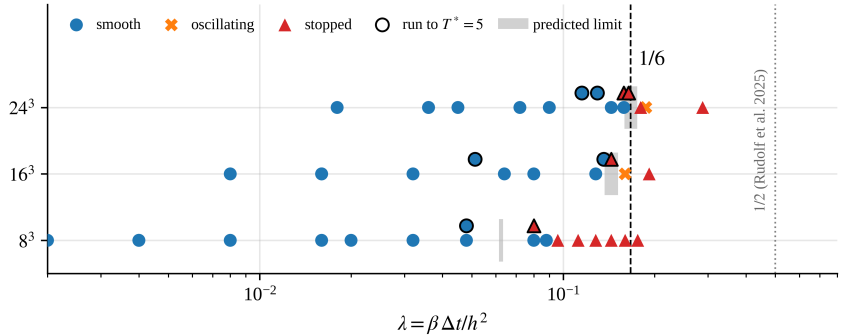
$$(\eta_{\phi,i} + \eta_i \psi_i^2)\,\dot{\phi}_i + \eta_i \phi_i \psi_i\,\dot{\psi}_i = c^* \psi_i \textstyle\sum_j A_{ij} \phi_j \psi_j - \gamma - \text{(barrier)}$$
$$\eta_i\,(\phi_i \psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) = c^* \phi_i \textstyle\sum_j A_{ij} \phi_j \psi_j - \alpha^* b_i\,\psi_i - \text{(barrier)}$$

with $\sum_i \phi_i + \phi_0 = 1$; the barrier keeps ϕ_i, ψ_i in $(0,1)$; implicit Euler and Newton, $\Delta t = 10^{-4}$.

One substep of the coupled ANSYS run

1. The element updates its field ϕ
2. At each Gauss point: $\alpha_{n+1} = \alpha_n + k_\alpha \phi\,\Delta t$, $\mathbf{F}_g = \alpha \mathbf{I}$, stress and tangent
3. Where $\phi_{3D} \geq \phi_{\min}$: advance the point model by $s\,\Delta t$ ($s = 0.1$) with $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$, linear in time; store ϕ_i, ψ_i and $\sum_i \bar{\phi}_i$
4. The element advances ϕ_{3D} (Eq. 34 with the front term $-r \frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$), c and a, consumed by $g\,\phi_{3D}(\sum_i \bar{\phi}_i)$

Appendix D: stability of the explicit field update



$\lambda = \beta\Delta t/h^2$; grey: the limit from a Bloch analysis of the NEM stencil on the lattice of the Gauss points (thesis, Appendix D). $h^2/(2\Delta t) \geq \beta$ of Rudolf et al. 2025 is the one-dimensional limit $(1/2)$; the seven-point stencil gives $1/6$; the NEM fit with its weight width $\beta^* = 0.2$ mm fixed in millimetres gives 0.06 on 8^3 and, depending on which equidistant points enter the 30 neighbours, 0.12–0.14 on 16^3 and 0.12–0.16 on 24^3. Runs to $T^* = 5$ (black edge) confirm it; runs to $T^* = 1.1$ are too short to show an instability near the limit.

Backup 1: stationarity conditions of the point model

$\dot{\Psi} + \Delta + C \to$ stat with respect to $\dot{\phi}_i$, $\dot{\psi}_i$, $\dot{\phi}_0$ and γ, where

$$\Psi = -\tfrac{1}{2} c^* \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}} + \tfrac{1}{2} \alpha^* \boldsymbol{\psi}^{\mathrm{T}} \mathbf{B}\, \boldsymbol{\psi}, \qquad \Delta = \tfrac{1}{2} \dot{\bar{\boldsymbol{\phi}}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2} \dot{\boldsymbol{\phi}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\boldsymbol{\phi}}, \qquad C = \gamma\Big(\sum_{l=0}^{n} \phi_l - 1\Big)$$

gives, with $\bar{\phi}_i = \phi_i \psi_i$ and $\dot{\bar{\phi}}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$,

$$\begin{aligned}
0 &= -c^* \psi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i \big(\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i\big) + \gamma, \\
0 &= -c^* \phi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \alpha^* b_i \psi_i + \eta_i \big(\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i\big), \\
0 &= \gamma + \dot{\phi}_0, \qquad 0 = \textstyle\sum_{l=0}^{n} \phi_l - 1.
\end{aligned}$$

- $2n+2$ equations; implicit Euler and Newton, $\Delta t = 10^{-4}$; a logarithmic barrier keeps ϕ_i, ψ_i, ϕ_0 in $(0,1)$
- The second term of Δ is needed: the first alone gives a singular system
- Only the symmetric part of $\mathbf{A}$ enters $\Psi \Rightarrow \mathbf{A} = \mathbf{A}^{\mathrm{T}}$

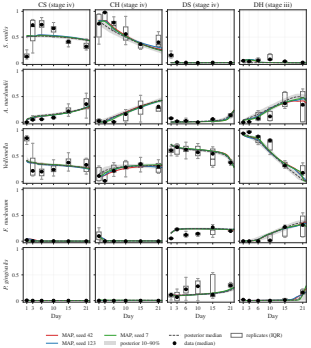
Backup 2: symmetric interactions compared with an asymmetric model

	gLV (L-BFGS-B)	Hamilton TMCMC
interaction matrix	asymmetric, 25	symmetric, 15
parameters	30	20
RMSE (four conditions)	0.012–0.032	0.027–0.179
uncertainty	point estimate	posterior, three seeds
$\sum_i \phi_i = 1$	enforced afterwards	exact

- The asymmetric model fits better, but has more parameters (30) than observations (25); its out-of-sample error was not assessed
- Directed effects (one species feeding another) cannot be represented; stated as a limitation

$\rightarrow$ Symmetry costs accuracy of the fit, but gives a posterior and exact conservation.

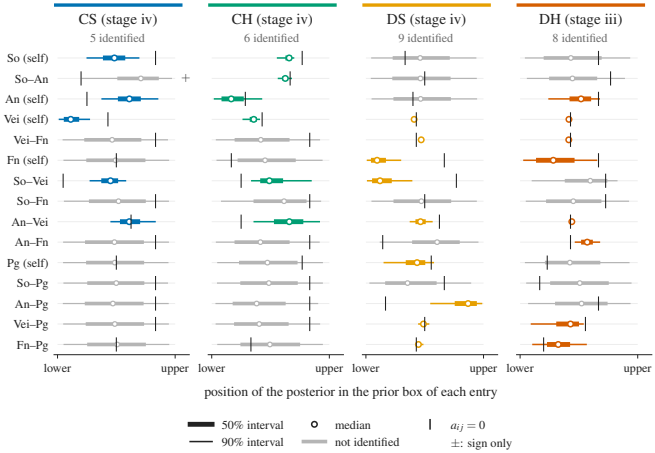
Backup 3: four stages and three seeds



Stages: (i) composition; (ii) narrowed box; (iii) with the live fraction, full box; (iv) narrowed box. Accepted only if recomputed likelihoods agree, tempering reaches $\beta = 1$, every particle moves enough, and the seeds agree in $\max \ln \mathscr{L}$ and in the medians.

$\rightarrow$ Several runs that looked converged failed these checks and were repeated; DS and DH are from stage (iii).

Backup 4: posterior distributions of the interactions



- Marginal posteriors of the 15 entries, drawn by their position in the prior box
- Identified: posterior SD below 0.8 of a uniform distribution on the box
- Sign only: 90 % interval excludes zero

Backup 5: sensitivity of the coupled examples

- **Time step** (outer step $0.025 \to 0.0025$): composition and $\alpha - 1$ within 3 and 0.3 %; stress $-6\,\%$ (separate) and $+15\,\%$ (mixed) at 0.025, so the stress is taken at the small step
- **Start values of new points**: neighbours' living composition against the seed composition, a few percent in the cube; in the maze of Klempt et al. 2024 recovery after the antibiotic (0.35) or not (0.04)
- **Antibiotic over time** (constant, pulses, sine, same dose): growth 0.1 %, stress 0.8 %; the living fraction follows the dose (pulsed: recovers $0.17 \to 0.30$)
- **Nutrient diffusivity**: with $\dot{c}$ and d of Table 2 the fields adjust in $4 \times 10^{-8}\,T^*$; only a value eight orders smaller changes the result
- **Stress mesh**: 16^3 against 64^3 elements, 0.2 %

$\to$ The composition is robust; the stress depends on the time step and on how the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$ is discretised.

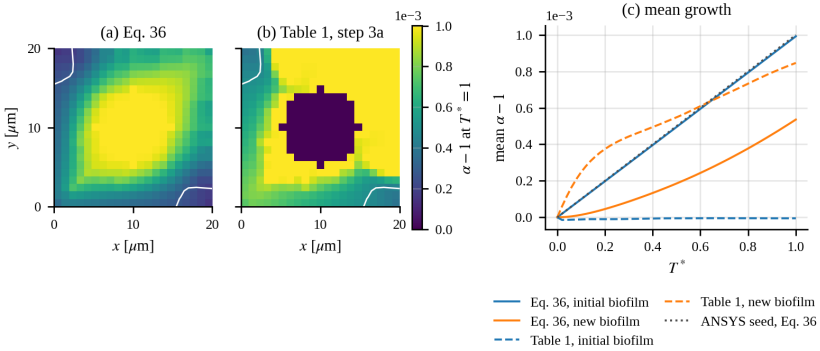
Backup 6: sensitivity to the diffusion coefficient β

Mean stress in the seed at $T^* = 1.1$ [Pa]

β [mm$^2/T^*$]	0.005	0.01	0.02	0.05	0.1
8^3	-2.1×10^{-4}	-2.7×10^{-4}	-2.5×10^{-4}	-1.3×10^{-4}	-5.6×10^{-5}
16^3	-3.9×10^{-4}	-3.4×10^{-4}	-2.5×10^{-4}	$-1.2 \times 10^{-4\,\dagger}$	$-3.8 \times 10^{-5\,\dagger}$

- $\beta = 0.02$: Table 2 of Klempt et al. 2024 ($\beta = 2$) converted to the 2 mm cube (assumption)
- ϕ diffusing out of the seed lowers its growth near the surface; the step in α between inside and surface loads the seed
- For $\beta \geq 0.02$ the diffusion length $\sqrt{\beta T^*} \geq 0.14$ mm is resolved by the mesh
- † before the correction of the surface points

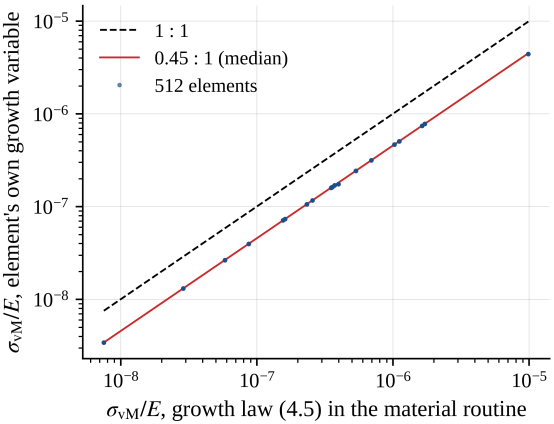
Backup 7: the two growth laws of Klempt et al. 2024



The paper gives $\dot{\alpha} = k_\alpha \phi$ (Eq. 36) in the text and a different update in Table 1, step 3a. On the reproduced field of test case 4.1 they differ: a seed full from the start grows only under Eq. 36.

→ I follow Eq. 36, as the text of the paper and the element do.

Backup 8: Eq. 36 against the element's own growth variable



- Whole model, 512 elements: the two von Mises fields have the same pattern, cosine similarity 1.0000
- The element's variable gives 0.455 times the stress of Eq. 36, which is $\frac{1}{2} \times \frac{10}{11}$: the mean over both species fields, one of them unused, and a lag of one substep
- With Eq. 36 at the Gauss points, the growth is counted once per increment

$\rightarrow$ Same spatial pattern; the difference in magnitude is fully explained by how the element's variable is formed, not by the mechanics.

Backup 9: mesh and incompressibility

- Python, same mechanical problem on 8^3, 16^3, 32^3 linear hexahedra: seed averages on 8^3 are 1.6–1.75 times those on 16^3 and 32^3, which agree with each other; corner peaks do not converge (singular corners) $\Rightarrow$ seed averages are quoted, not peaks
- Mean-dilatation (B-bar) integration, as SOLID185 with KEYOPT(2)= 0: mean stress -3.8 and -4.1×10^{-5} Pa for $\nu = 0.49$ and 0.499 on 8^3 $\Rightarrow$ no volumetric locking
- Full $2 \times 2 \times 2$ integration would overestimate the mean stress by a factor of 3.7 on 8^3 and 2.3 on 16^3 at $\nu = 0.49$
- ANSYS: 16^3 and 24^3 with every boundary condition mapped by position; 32^3 exceeds the memory of the workstation

$\rightarrow$ The quoted seed averages are mesh-converged on 16^3 and 24^3; B-bar avoids locking at $\nu = 0.49$.

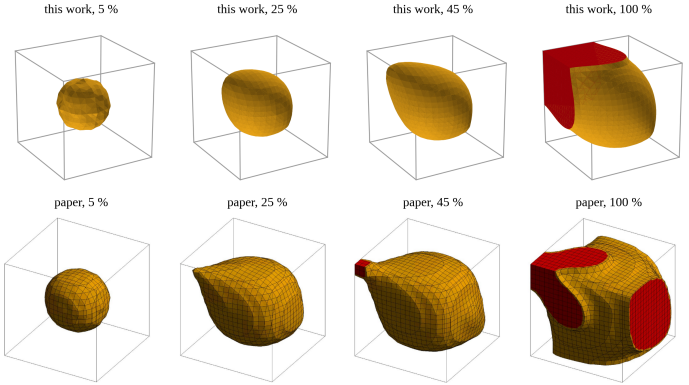
Backup 10: correction of the surface points of the element

- In the element, ϕ at the surface points was not updated and stayed 0: every face acted as $\phi = 0$, where the model has zero flux
- Corrected: each surface point takes the distance-weighted mean of its neighbouring Gauss points; all thesis runs repeated

runs	seed von Mises	seed pressure	$\alpha - 1$
$\beta = 0.02$ (8^3, 16^3, one and two species)	$\leq 0.01\,\%$	$\leq 0.13\,\%$	unchanged
$\beta = 0.05$, 8^3	$+0.3\,\%$	$-3.5\,\%$	$0.02\,\%$
$\beta = 0.1$, 8^3	$+6.7\,\%$	$-30\,\%$	$0.5\,\%$

$\rightarrow$ The main results change by less than $0.5\,\%$; only runs in which ϕ reaches the faces change noticeably, and their corrected values are used.

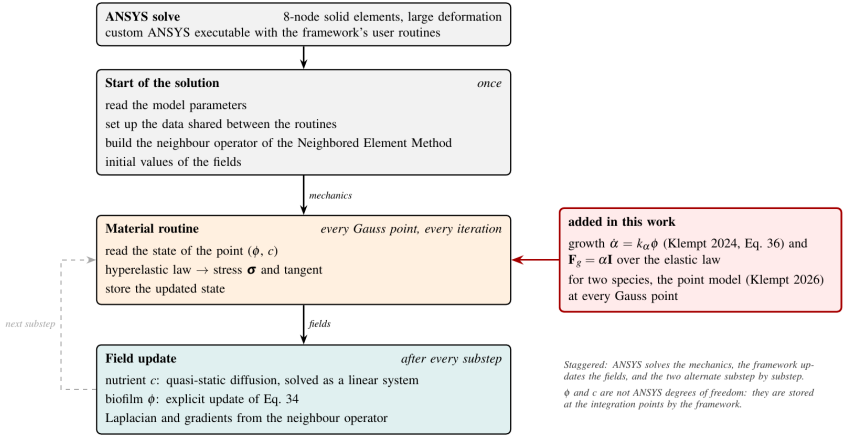
Backup 11: reproduction of Klempt et al. 2024, test case 4.1



Python reproduction with the parameters of Table 2; consumption $g\phi c$ as in the authors' implementation (printed: $g\phi$), front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$ with an isotropic share.

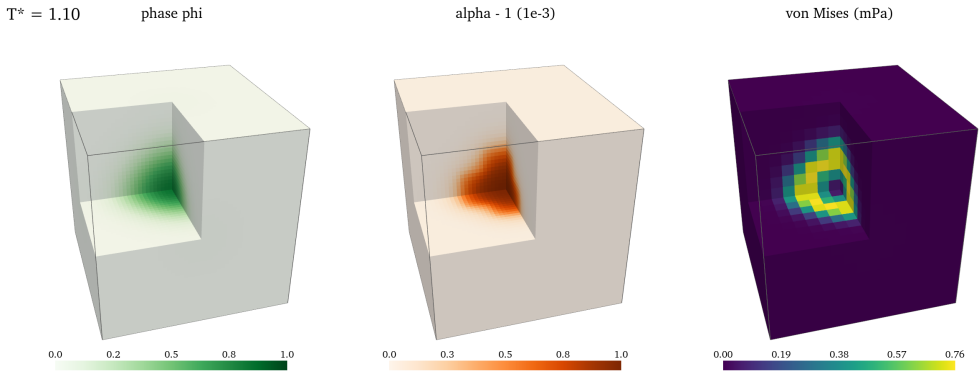
→ Case 4.1 is reproduced with stated departures from the printed equations (thesis, Appendix C).

Backup 12: solution loop of the element framework



→ The material routine is called at every Gauss point and iteration; the fields ϕ and c are advanced after each substep.

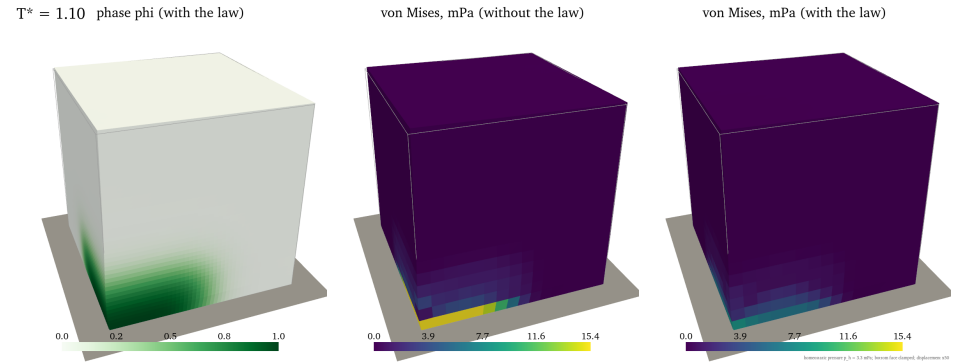
Backup 13: the verification problem in 3D



Seed in the centre of the 2 mm cube, one octant cut away; $\beta = 0.02\,\mathrm{mm}^2/T^*$, $k_\alpha = 10^{-3}/T^*$, $E = 10\,\mathrm{Pa}$. Field: reference solution on 48^3; stress: 16^3. Video: `assets/anim_wp3_seed_3d.mp4`.

$\rightarrow$ Seed von Mises 6.19×10^{-4} Pa at $T^* = 1.1$, within 0.2 % of the value in the thesis; the stress forms a ring at the edge of the seed.

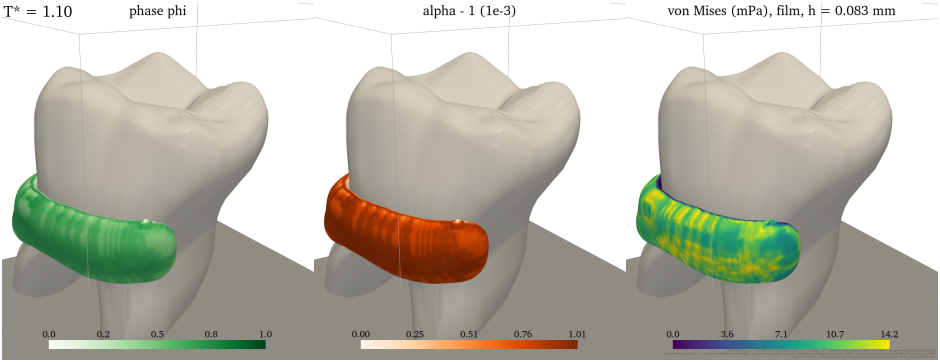
Backup 14: outlook, stress at a wall and a homeostatic-pressure law



Not part of the thesis: the same model with the seed in a corner on a clamped face, without and with growth scaled by $g = \min(1, \max(0, 1 - p/p_h))$; $p_h = 3.3$ MPa chosen by me. Video: `assets/anim_law_3d.mp4`.

$\rightarrow$ In the biofilm next to a fixed surface the stress is 10 to 12 times the seed stress of the centre case (band mean, converged to 1 %); limiting growth by pressure removes the concentration. Continuation at Keio.

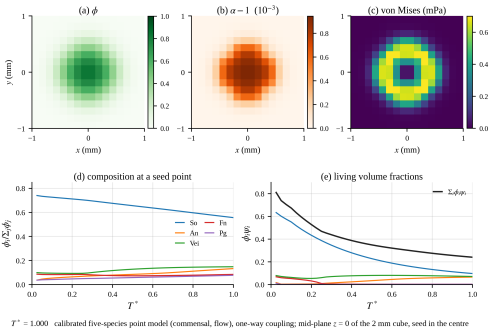
Backup 15: outlook, a biofilm on a real molar



Not part of the thesis: the same growth model on a lower first molar at real size (public surface model, CC BY 4.0), a 0.5 mm film near the gum line. Resolved with six elements across the film, the film mean von Mises stress is 3.3 mPa ($\pm 1\,\%$), highest in the 0.1 mm next to the tooth.

$\rightarrow$ Thin films on real teeth need a film-resolving mesh; with it the stresses converge. Continuation at Keio.

Backup 16: the calibrated five-species model in the element, over time

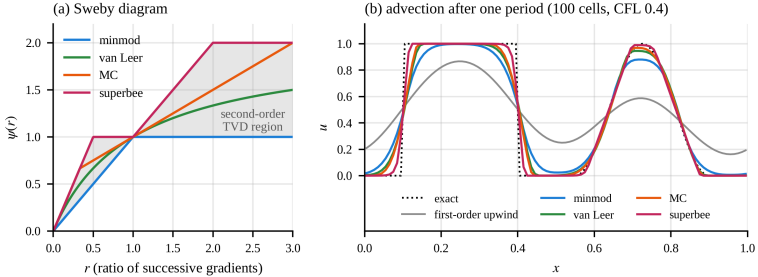


$T^* = 1.000$ calibrated five-species point model (commensal, flow), one-way coupling; mid-plane $z = 0$ of the 2 mm cube, seed in the centre

The run of Section 4 (commensal flow condition, final MAP, Day-1 start, one-way coupling) on the finer 16^3 mesh, 40 output times up to $T^* = 1$. At $T^* = 1$: seed mean $\alpha - 1 = 6.56 \times 10^{-4}$, von Mises 5.74×10^{-4} Pa (8^3 in the thesis: 6.75×10^{-4}, 4.68×10^{-4} Pa); shares within 0.01 of the 8^3 run. Video: `assets/anim_cal5_ch.mp4`.

→ *S. oralis* falls from 0.74 to 0.56 and the living fraction from 0.81 to 0.24, while growth and stress do not depend on the composition (one-way coupling).

Backup 17: slope limiters for the front term



(a) Limiter $\psi(r)$ against the ratio r of successive gradients; grey: the second-order TVD region of Sweby (1984), bounded by minmod below and superbee above. (b) Textbook test, not the biofilm model: $u_t + u_x = 0$, 100 cells, MUSCL reconstruction, upwind flux, SSP Runge–Kutta 3, CFL 0.4, one period. Minmod smears the edges and clips the smooth peak to 0.88; superbee keeps the edges sharpest and slightly steepens the smooth peak.

$\rightarrow$ The front term $-r\frac{c}{k+c}\mathbf{n}_c \cdot \nabla\phi$ moves a sharp edge; superbee keeps it sharp, so the mixed stress converges on coarser grids than with minmod.